

Pico Gym Exclusive - Bit-O-Asm-3

Bit-O-Asm-3

Reverse Engineering

Medium

100 pts

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Can you figure out what is in the `eax` register? Put your answer in the picoCTF flag format: `picoCTF{n}` where `n` is the contents of the `eax` register in the decimal number base. If the answer was `0x11` your flag would be `picoCTF{17}`.

In this challenge, we're asked to look at an assembly dump for a program and figure out what's in the `eax` register at the end of the program

Pico Gym Exclusive – Bit-O-Asm-3 x86 Assembly Instructions

But to understand the contents of the disassembly dump file, we'll need to understand a bit about the x86 assembly language



Pico Gym Exclusive – Bit-O-Asm-3 x86 Assembly Instructions

Assembly languages are the programming languages closest to the pure machine code that computer processors execute



Pico Gym Exclusive – Bit-O-Asm-3 x86 Instructions

```
imul    eax,DWORD PTR [rbp-0x8]  
add     eax,0x1f5  
mov     DWORD PTR [rbp-0x4],eax
```

To solve this challenge, we need to understand three x86 assembly instructions: MOV, ADD, and IMUL

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The MOV Instruction

```
mov    DWORD PTR [rbp-0x8],0x4
```

MOV <location affected>, <value>

MOV is an assembly instruction which moves data into memory locations

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The MOV Instruction

```
mov    DWORD PTR [rbp-0x8],0x4
```

MOV <location affected>, <value>

In this example, the value 0x4 (decimal 4) is moved into the DWORD PTR [rbp-0x8] location

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The ADD Instruction

```
add    eax,0x1f5
```

ADD <location affected>, <value>

The ADD instruction is used to add a value to the current value at a specific location

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The ADD Instruction

```
add    eax,0x1f5
```

ADD <location affected>, <value>

In this example, the value 0x1f5 (decimal 501) is added to the value at the EAX location

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The IMUL Instruction

```
imul    eax,DWORD PTR [rbp-0x8]
```

IMUL <location affected>, <value>

The IMUL instruction multiplies the value of a specified location by value

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The IMUL Instruction

```
imul    eax,DWORD PTR [rbp-0x8]
```

IMUL <location affected>, <value>

In this example, the value of the EAX location is multiplied by the value at the DWORD PTR [rbp-0x8] location

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Keeping Track of Memory Locations

Location			Value
EAX			0
DWORD	PTR	[rbp-0x4]	0
DWORD	PTR	[rbp-0x8]	0
DWORD	PTR	[rbp-0xc]	0

We will need to track the contents of the memory locations illustrated above

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First Instruction

```
mov     DWORD PTR [rbp-0xc],0x9fe1a  
; rbp-0xc now holds 654874
```

EAX = 0

RBP-0x4 = 0

RBP-0x8 = 0

RBP-0xC = 654874

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Second Instruction

```
mov     DWORD PTR [rbp-0x8],0x4  
; rbp-0x8 now holds 4
```

EAX = 0

RBP-0x4 = 0

RBP-0x8 = 4

RBP-0xC = 654874

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Third Instruction

```
mov     eax,DWORD PTR [rbp-0xc]  
; eax now holds 654874
```

EAX = 654874

RBP-0x4 = 0

RBP-0x8 = 4

RBP-0xC = 654874

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Fourth Instruction

```
imul    eax,DWORD PTR [rbp-0x8]  
; eax now holds 2619496
```

EAX = 2619496

RBP-0x4 = 0

RBP-0x8 = 4

RBP-0xC = 654874

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Fifth Instruction

```
add    eax,0x1f5  
; eax now holds 2619997
```

EAX = 2619997

RBP-0x4 = 0

RBP-0x8 = 4

RBP-0xC = 654874

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Sixth Instruction

```
mov     DWORD PTR [rbp-0x4],eax  
; rbp-0x4 now holds 2619997
```

EAX = 2619997

RBP-0x4 = 2619997

RBP-0x8 = 4

RBP-0xC = 654874

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Seventh Instruction

```
mov     eax,DWORD PTR [rbp-0x4]  
; eax now holds 2619997
```

EAX = 2619997

RBP-0x4 = 2619997

RBP-0x8 = 4

RBP-0xC = 654874